

Imperfect Information & Global Sensitivity Analysis

Lecture

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Draft Material — This content is under development and subject to change.

1 Recap: Monday's Key Results

On Monday we introduced the umbrella problem with $C = 5$, $L = 50$, and $p = 0.30$.

	Bring umbrella	Don't bring
Rain ($p = 0.3$)	-5	-50
No rain ($1 - p = 0.7$)	-5	0

We computed:

- **Without information:** bring the umbrella, expected cost = \$5.00
- **With perfect information:** expected cost = $0.3 \times 5 + 0.7 \times 0 = \1.50
- **EVPI** = $5.00 - 1.50 = \$3.50$

The EVPI tells you the maximum you'd pay for a perfect oracle. But no real forecast is perfect. Today we ask: what if the forecast is noisy?

2 Value of Imperfect Information

2.1 An Imperfect Rain Forecast

Suppose a weather app gives you a rain forecast each morning. It's not perfect:

- When it actually rains, the app says "rain" 90% of the time (and misses 10%)
- When it doesn't rain, the app says "no rain" 80% of the time (but falsely says "rain" 20%)

In probabilistic notation:

- $P(\text{forecast rain} \mid \text{rain}) = 0.9$ — the **hit rate** (sensitivity)
- $P(\text{forecast no rain} \mid \text{no rain}) = 0.8$ — the **correct rejection rate** (specificity)

```
# Parameters from Monday's umbrella problem
C = 5      # cost of bringing umbrella
L = 50     # loss if unprotected and it rains
```

```

p = 0.30 # base rate of rain

# Forecast quality
hit_rate = 0.90 # P(forecast rain | rain)
correct_rejection = 0.80 # P(forecast no rain | no rain)
false_alarm_rate = 1 - correct_rejection # P(forecast rain | no rain)
miss_rate = 1 - hit_rate # P(forecast no rain | rain)

```

Using the base rate $p = 0.3$, we can compute the probability of each forecast:

$$P(\text{forecast rain}) = P(\text{forecast rain} \mid \text{rain}) \cdot p + P(\text{forecast rain} \mid \text{no rain}) \cdot (1 - p) \quad (1)$$

```

p_forecast_rain = hit_rate * p + false_alarm_rate * (1 - p)

```

```

0.41

```

Similarly, $P(\text{forecast no rain}) = P(\text{forecast no rain} \mid \text{rain}) \cdot p + P(\text{forecast no rain} \mid \text{no rain}) \cdot (1 - p)$:

```

p_forecast_no_rain = miss_rate * p + correct_rejection * (1 - p)

```

```

0.59

```

And the posterior probabilities of rain given each forecast (via Bayes' rule):

$$P(\text{rain} \mid \text{forecast rain}) = \frac{P(\text{forecast rain} \mid \text{rain}) \cdot P(\text{rain})}{P(\text{forecast rain})} \quad (2)$$

```

p_rain_given_forecast_rain = hit_rate * p / p_forecast_rain

```

```

0.6585365853658537

```

And $P(\text{rain} \mid \text{forecast no rain}) = \frac{P(\text{forecast no rain} \mid \text{rain}) \cdot P(\text{rain})}{P(\text{forecast no rain})}$:

```

p_rain_given_forecast_no_rain = miss_rate * p / p_forecast_no_rain

```

```

0.050847457627118633

```

The decision rule from Monday: bring the umbrella when $P(\text{rain} \mid \text{forecast}) \times L > C$.

When the app says “rain”, the expected loss without protection is $P(\text{rain} \mid \text{forecast rain}) \times L$:

```

p_rain_given_forecast_rain * L

```

```

32.926829268292686

```

This exceeds $C = 5$, so **bring the umbrella**. When the app says “no rain”, it’s $P(\text{rain} \mid \text{forecast no rain}) \times L$:

```
p_rain_given_forecast_no_rain * L
```

```
2.5423728813559316
```

This is well below $C = 5$, so **leave it**.

2.2 Computing Expected Cost with the Imperfect Forecast

Now we can compute the expected cost of following the app’s advice. There are four possible outcomes:

Forecast	Actual	Action	Cost	Probability
Rain	Rain	Bring	C	hit rate $\times p$
Rain	No rain	Bring	C	false alarm rate $\times (1 - p)$
No rain	Rain	Leave	L	miss rate $\times p$
No rain	No rain	Leave	0	correct rejection $\times (1 - p)$

The expected cost is $E_F = \sum_i \text{cost}_i \times \text{probability}_i$:

```
E_F = (hit_rate * p) * C +
      (false_alarm_rate * (1 - p)) * C +
      (miss_rate * p) * L +
      (correct_rejection * (1 - p)) * 0
```

```
3.5499999999999994
```

Compare to our benchmarks. Without information, the best strategy is to always bring the umbrella (since $pL > C$), at cost $E_C = C$. With perfect information, you bring it only when it rains, at cost $E_P = p \times C$:

```
E_no_info = C          # always bring umbrella
E_perfect = p * C      # bring only when it rains

println("No information:      \$$E_no_info")
println("Imperfect forecast: \$$ (round(E_F; digits=2))")
println("Perfect information: \$$E_perfect")
```

```
No information:      $5
Imperfect forecast:  $3.55
Perfect information:  $1.5
```

The **expected value of imperfect information** is $EVII = E_C - E_F$, and for comparison $EVPI = E_C - E_P$:

```
EVPI = E_no_info - E_perfect
EVII = E_no_info - E_F

println("EVPI = \$$EVPI")
println("EVII = \$$ (round(EVII; digits=2))")
println("Forecast captures $(round(100 * EVII / EVPI; digits=0))% of perfect info value")
```

```
EVPI = $3.5
EVII = $1.45
Forecast captures 41.0% of perfect info value
```

2.3 The General Formula

The umbrella example illustrates a general pattern. To compute the value of an imperfect information source, think through three steps:

1. **Consider all the signals you might receive.** The weather app might say “rain” or “no rain.” A climate model might project high or low sea level rise. Each possible signal z has some probability $p(z)$ of occurring.
2. **For each signal, ask: if I got this signal, what would I do, and how good is that in expectation?** Receiving signal z updates your beliefs about the world via Bayes’ rule, giving posterior $p(s | z)$. You then choose the action that maximizes expected payoff under those updated beliefs: $\max_a \int U(a, s) p(s | z) ds$.
3. **Average over all the signals you might get.** You don’t know in advance which signal you’ll receive, so take the expectation over z .

The EVII is the difference between this average and what you’d do with no signal at all:

$$EVII = \underbrace{\int \left[\max_a \int U(a, s) p(s | z) ds \right] p(z) dz}_{\text{get signal, update, optimize, average}} - \underbrace{\max_a \int U(a, s) p(s) ds}_{\text{optimize with prior alone}} \quad (3)$$

This reduces to EVPI when the signal perfectly reveals s (the posterior is a point mass — you know for certain). And it reduces to zero when the signal is pure noise ($p(s | z) = p(s)$ — you learn nothing).

2.4 Blackwell’s Theorem

A key result [1]: $0 \leq EVII \leq EVPI$. An imperfect forecast can never hurt you in expectation — because you can always ignore it and revert to your prior decision.

But the relationship between forecast quality and value is nonlinear. Forecasts are most valuable near the **decision boundary**, when $C/L \approx p$. When the decision is already obvious (the base rate is far from the threshold), even a much better forecast has little value.

3 Global Sensitivity Analysis

We now turn to a complementary tool. VOI asks: *which uncertainty, if resolved, would most change my decision?* Global sensitivity analysis (GSA) asks a different question: *which uncertainty contributes most to the variability in my output?*

These are **not the same question**, and the answers can disagree. A parameter can dominate the output variance but have zero EVPI — if it doesn't change which action is best. Both are useful, and understanding the distinction is essential.

3.1 What Is Sensitivity Analysis?

Sensitivity Analysis (SA) investigates how the variation in the output of a model can be attributed to variations of its input factors [2]. In our notation, we have a model $Y = g(X_1, X_2, \dots, X_M)$ where the X_i are uncertain inputs and Y is a scalar output of interest.

SA addresses questions like:

- Which input factors cause the largest variation in the output?
- Are there inputs whose variability has a negligible effect?
- Are there **interactions** — does the effect of one input depend on the value of another?

3.2 Three Purposes of Sensitivity Analysis

F. Pianosi *et al.* [2] distinguish three purposes for SA, each guiding the choice of method:

1. **Screening** (factor fixing): identify which inputs have a *negligible* influence on the output, so they can be fixed to simplify the model. “Which inputs can I safely ignore?”
2. **Ranking** (factor prioritization): rank inputs by their relative contribution to output variability, so you know where to focus research or data collection. “Which input matters most?”
3. **Mapping**: identify regions of the input space that produce extreme or otherwise interesting output values. “What combinations of inputs lead to failure?”

These purposes are complementary, not exclusive. A typical analysis might start with screening (cheap methods to eliminate unimportant inputs), then ranking (more expensive methods on the remaining inputs), and finally mapping (to understand failure modes).

3.3 Local Sensitivity Analysis

The simplest approach is **one-at-a-time** (OAT) sensitivity analysis: vary one input while holding all others fixed at baseline values, and see how the output changes. The local sensitivity of the output to input x_i is the partial derivative:

$$S_i^{\text{local}} \approx \left. \frac{\partial g}{\partial x_i} \right|_{\mathbf{x}=\bar{\mathbf{x}}} \quad (4)$$

In practice, we approximate Equation 4 using a **finite difference**. Pick a baseline $\bar{\mathbf{x}}$, perturb input x_i by a small amount Δ_i , and compute:

$$\hat{S}_i = \frac{g(\bar{x}_1, \dots, \bar{x}_i + \Delta_i, \dots, \bar{x}_M) - g(\bar{x}_1, \dots, \bar{x}_i, \dots, \bar{x}_M)}{\Delta_i} \quad (5)$$

This requires only $M + 1$ model evaluations: one at the baseline, plus one perturbation per input. It's fast, intuitive, and often the first thing analysts try.

Of course, Equation 5 only gives the sensitivity at a single baseline. A natural extension is to repeat the same finite difference from multiple starting points $\bar{x}^1, \bar{x}^2, \dots, \bar{x}^r$ and summarize the results — for instance by averaging the absolute values or examining their spread. This is still OAT (one input perturbed at a time), but it gives a more global picture of the sensitivity landscape. That idea is exactly what the Morris method formalizes, as we'll see shortly.

This is what you did informally in Lab 5 when you optimized under RCP 8.5 and then under RCP 2.6 — one scenario at a time.

3.4 What Makes Sensitivity Analysis “Global”?

Local OAT analysis is fast, but it has serious limitations:

- It measures sensitivity at **one point** in input space — sensitivity may look completely different elsewhere
- It misses **interactions** between parameters
- The baseline choice $\bar{x}$ is arbitrary
- It does not explore the full range of each input

Global SA addresses all of these by varying **all inputs simultaneously** across their full ranges. F. Pianosi *et al.* [2] call this **All-At-a-Time** (AAT) sampling, in contrast to OAT.

Instead of asking “what happens when I wiggle x_i a little?”, GSA asks: “across the full distribution of inputs, how much of the output variability comes from x_i ?”

This captures:

- The full range of each input (not just a local derivative)
- Nonlinear effects
- Interactions between inputs

3.5 A Classification of SA Methods

F. Pianosi *et al.* [2] organize SA methods into five broad classes, ordered roughly by computational cost. The right method depends on your **purpose** (screening, ranking, or mapping) and your **budget** (how many model evaluations you can afford).

Method	Sampling	Cost (M inputs)	Best for	Key limitation
Perturbation / derivatives	OAT	$M + 1$	Quick local check	Local only, misses interactions
Elementary Effects (Morris)	OAT	$r(M + 1)$, $r \approx 10\text{--}50$	Screening	Approximate; detects interactions but can't identify specific pairs
Correlation / Regression (SRC)	AAT	Any MC sample	Ranking (linear models)	Assumes linearity or monotonicity

Method	Sampling	Cost (M inputs)	Best for	Key limitation
Regional SA (Monte Carlo filtering)	AAT	Any MC sample	Mapping	Requires output threshold
Variance-based (Sobol indices)	AAT	$> 1000M$	Ranking + interactions	Expensive; variance may miss skew
Density-based (PAWN, δ)	AAT	$> 1000M$	Ranking (non-standard outputs)	Most expensive

We'll look at two of these methods in detail: one OAT method (Morris) and one AAT method (Sobol' indices).

3.5.1 The Elementary Effects Test (Morris Method)

The local finite difference we computed earlier gives the sensitivity at one point. Morris's key idea: **do that same local calculation from many different starting points**, then summarize.

The method builds r random trajectories through the input space. Each trajectory has $M + 1$ points, varying one input at a time but starting from a different randomly-selected location each time. For each input x_i and trajectory j , it computes the **elementary effect**:

$$EE_i^j = \frac{g(\bar{x}_1^j, \dots, \bar{x}_i^j + \Delta_i, \dots, \bar{x}_M^j) - g(\bar{x}_1^j, \dots, \bar{x}_i^j, \dots, \bar{x}_M^j)}{\Delta_i} \quad (6)$$

This is the same finite difference as Equation 5 — but computed at many locations across the input space. Then it summarizes across trajectories:

- The **mean** of $|EE_i|$: measures overall influence (useful for ranking)
- The **standard deviation** of EE_i : high SD means the sensitivity *changes* across the input space, indicating interactions or nonlinearity

A plot of mean vs. standard deviation for each input tells the whole story:

- Near the origin: unimportant inputs (screen these out)
- Far right, low SD: strong main effect, few interactions
- Far right, high SD: important and interacting with other inputs

The total cost is $r(M + 1)$ model evaluations. With $r = 10$ and $M = 5$ inputs, that's just 60 model runs — feasible even for expensive models.

3.6 Variance Decomposition and Sobol Indices

For a model $Y = g(X_1, X_2, \dots, X_k)$ with independent inputs, the total variance decomposes as:

$$\text{Var}(Y) = \sum_i V_i + \sum_{i < j} V_{ij} + \dots + V_{12\dots k} \quad (7)$$

where:

- $V_i = \text{Var}(\mathbb{E}[Y | X_i])$ is the variance due to X_i alone
- V_{ij} captures the interaction between X_i and X_j

- Higher-order terms capture three-way and higher interactions

The **first-order Sobol index** is the fraction of output variance due to X_i alone:

$$S_i = \frac{V_i}{\text{Var}(Y)} = \frac{\text{Var}(\mathbb{E}[Y | X_i])}{\text{Var}(Y)} \quad (8)$$

The **total-order Sobol index** captures all variance involving X_i , including all interactions:

$$S_{T_i} = 1 - \frac{\text{Var}(\mathbb{E}[Y | X_{\sim i}])}{\text{Var}(Y)} \quad (9)$$

where $X_{\sim i}$ means all inputs *except* X_i .

Interpreting these indices:

Pattern	Interpretation
S_i large, $S_{T_i} \approx S_i$	X_i has a strong main effect, few interactions
S_i small, S_{T_i} large	X_i matters mainly through interactions
S_i small, S_{T_i} small	X_i doesn't matter much
$\sum S_i \approx 1$	Model is mostly additive

A useful rule of thumb from F. Pianosi *et al.* [2]: the total-order index $S_{T_i} \approx 0$ is a necessary and sufficient condition for screening out an unimportant factor. First-order indices are better suited for ranking, though they can understate importance when interactions are strong.

Note that variance is not the only way to define “sensitivity.” Density-based methods (e.g., PAWN) measure sensitivity through changes in the full output distribution rather than just its variance, which can be more informative when the output is skewed or multimodal. But variance decomposition is the most widely used framework, so we focus on it here.

3.7 Reading a Sensitivity Analysis

When you encounter a sensitivity analysis — or conduct one yourself — there are several things you need to understand before trusting the results [2].

3.7.1 What inputs were varied, and what was held fixed?

Every SA starts with choices about scope: which inputs are treated as uncertain, which are fixed at default values, and what scalar output is being analyzed. These choices frame the entire analysis. A different choice of output (expected annual damages vs. optimal dike height) can produce a completely different ranking of inputs.

3.7.2 What distributions were assumed?

Every GSA calculation requires a probability distribution over the inputs. The expectations $\mathbb{E}[Y | X_i]$ and variance $\text{Var}(Y)$ in Equation 8 and Equation 9 are only defined relative to a joint distribution $p(X_1, \dots, X_k)$. Change the distribution, change the Sobol indices.

This is not a minor technical detail. If you use Uniform $(-\pi, \pi)$ for all inputs, you get one set of Sobol indices. If you concentrate one input around a narrow range using a Normal distribution,

its first-order index may drop to near zero — not because the model is less sensitive to it, but because you’ve assumed it doesn’t vary much.

As F. Pianosi *et al.* [2] emphasize, the definition of the input variability space is “one of the most delicate steps in the application of GSA.” Many software packages only ask for ranges, which implicitly assumes uniform distributions. That’s a strong assumption, and it should be a deliberate choice, not a default.

3.7.3 Are the results converged?

Sensitivity indices computed from a finite sample are *estimates*, not exact values. Two checks matter:

- **Convergence:** do the indices stabilize as you increase the sample size? If the ranking is still shifting, you don’t have enough samples.
- **Robustness:** do you get similar indices from different sub-samples (bootstrapping)? A Sobol index of 0.35 ± 0.20 tells a very different story than 0.35 ± 0.02 .

When results are unexpected — a parameter you thought was unimportant turns out to dominate — that’s a signal to investigate, not dismiss.

4 Summary

Today we covered two complementary tools:

1. **Value of imperfect information (EVII):** consider all the signals you might get, figure out what you’d do for each, and average. An imperfect but calibrated forecast always has non-negative value (Blackwell’s theorem).
2. **Global sensitivity analysis:** a family of methods for understanding which inputs drive output variability. Methods range from cheap screening tools (Morris) to comprehensive variance decomposition (Sobol indices). When reading or conducting a SA, always ask: what inputs were varied, what distributions were assumed, and are the results converged?

Both tools share a critical requirement: **probability distributions over the uncertain inputs**. When you can’t specify those distributions confidently, you’re in the territory of **deep uncertainty** — which is where we’re headed in Week 8.

4.1 References

Bibliography

- [1] D. Blackwell, “Equivalent Comparisons of Experiments,” *The Annals of Mathematical Statistics*, vol. 24, no. 2, pp. 265–272, 1953, doi: 10.1214/aoms/1177729032.
- [2] F. Pianosi *et al.*, “Sensitivity Analysis of Environmental Models: A Systematic Review with Practical Workflow,” *Environmental Modelling & Software*, vol. 79, pp. 214–232, May 2016, doi: 10/f8n6zw.